



Inequalities and Feasible Choices

Grades 9-12 • SAT / ACT mathematics foundations

Algebra • Represent a constraint and identify feasible values.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Represent a constraint and identify feasible values.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



Two worked examples

Model 1. A club has at most \$135 to spend. Supplies cost \$45 plus \$7 per participant. What is the greatest whole number of participants it can support?

$45 + 7n \leq 135$; $n \leq 90/7$. The greatest whole number is 12. 13 participants would cost \$136, over the limit.

Model 2. A club has at most \$69 to spend. Supplies cost \$10 plus \$6 per participant. What is the greatest whole number of participants it can support?

$10 + 6n \leq 69$; $n \leq 59/6$. The greatest whole number is 9. 10 participants would cost \$70, over the limit.

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A club has at most \$63 to spend. Supplies cost \$40 plus \$2 per participant. What is the greatest whole number of participants it can support?

2. A club has at most \$123 to spend. Supplies cost \$25 plus \$9 per participant. What is the greatest whole number of participants it can support?

3. A club has at most \$156 to spend. Supplies cost \$45 plus \$8 per participant. What is the greatest whole number of participants it can support?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. A club has at most \$72 to spend. Supplies cost \$10 plus \$7 per participant. What is the greatest whole number of participants it can support?

5. A club has at most \$59 to spend. Supplies cost \$10 plus \$5 per participant. What is the greatest whole number of participants it can support?

6. A club has at most \$47 to spend. Supplies cost \$30 plus \$2 per participant. What is the greatest whole number of participants it can support?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. A club has at most \$99 to spend. Supplies cost \$40 plus \$5 per participant. What is the greatest whole number of participants it can support?

8. A club has at most \$62 to spend. Supplies cost \$15 plus \$6 per participant. What is the greatest whole number of participants it can support?

9. A club has at most \$141 to spend. Supplies cost \$25 plus \$9 per participant. What is the greatest whole number of participants it can support?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. A club has at most \$61 to spend. Supplies cost \$40 plus \$2 per participant. What is the greatest whole number of participants it can support?

11. A club has at most \$72 to spend. Supplies cost \$45 plus \$2 per participant. What is the greatest whole number of participants it can support?

12. A club has at most \$63 to spend. Supplies cost \$40 plus \$3 per participant. What is the greatest whole number of participants it can support?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. A club has at most \$96 to spend. Supplies cost \$25 plus \$6 per participant. What is the greatest whole number of participants it can support?

14. A club has at most \$75 to spend. Supplies cost \$40 plus \$3 per participant. What is the greatest whole number of participants it can support?

15. A club has at most \$152 to spend. Supplies cost \$45 plus \$9 per participant. What is the greatest whole number of participants it can support?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. A club has at most \$106 to spend. Supplies cost \$30 plus \$7 per participant. What is the greatest whole number of participants it can support?



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. A club has at most \$36 to spend. Supplies cost \$15 plus \$2 per participant. What is the greatest whole number of participants it can support?

2. A club has at most \$117 to spend. Supplies cost \$20 plus \$7 per participant. What is the greatest whole number of participants it can support?



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. A club has at most \$137 to spend. Supplies cost \$40 plus \$7 per participant. What is the greatest whole number of participants it can support?

4. A club has at most \$38 to spend. Supplies cost \$15 plus \$2 per participant. What is the greatest whole number of participants it can support?

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. A club has at most \$69 to spend. Supplies cost \$40 plus \$3 per participant. What is the greatest whole number of participants it can support?

6. A club has at most \$99 to spend. Supplies cost \$40 plus \$5 per participant. What is the greatest whole number of participants it can support?



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. A club has at most \$130 to spend. Supplies cost \$40 plus \$7 per participant. What is the greatest whole number of participants it can support?

2. A club has at most \$134 to spend. Supplies cost \$45 plus \$9 per participant. What is the greatest whole number of participants it can support?



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. A club has at most \$106 to spend. Supplies cost \$30 plus \$7 per participant. What is the greatest whole number of participants it can support?

4. A club has at most \$81 to spend. Supplies cost \$30 plus \$4 per participant. What is the greatest whole number of participants it can support?



Practice worked solutions

1. 11

$40 + 2n \leq 63$; $n \leq 23/2$. The greatest whole number is 11. 12 participants would cost \$64, over the limit.

2. 10

$25 + 9n \leq 123$; $n \leq 98/9$. The greatest whole number is 10. 11 participants would cost \$124, over the limit.

3. 13

$45 + 8n \leq 156$; $n \leq 111/8$. The greatest whole number is 13. 14 participants would cost \$157, over the limit.

4. 8

$10 + 7n \leq 72$; $n \leq 62/7$. The greatest whole number is 8. 9 participants would cost \$73, over the limit.

Practice worked solutions

5. 9

$10+5n \leq 59$; $n \leq 49/5$. The greatest whole number is 9. 10 participants would cost \$60, over the limit.

6. 8

$30+2n \leq 47$; $n \leq 17/2$. The greatest whole number is 8. 9 participants would cost \$48, over the limit.

7. 11

$40+5n \leq 99$; $n \leq 59/5$. The greatest whole number is 11. 12 participants would cost \$100, over the limit.

8. 7

$15+6n \leq 62$; $n \leq 47/6$. The greatest whole number is 7. 8 participants would cost \$63, over the limit.

Practice worked solutions

9. 12

$25 + 9n \leq 141$; $n \leq 116/9$. The greatest whole number is 12. 13 participants would cost \$142, over the limit.

10. 10

$40 + 2n \leq 61$; $n \leq 21/2$. The greatest whole number is 10. 11 participants would cost \$62, over the limit.

11. 13

$45 + 2n \leq 72$; $n \leq 27/2$. The greatest whole number is 13. 14 participants would cost \$73, over the limit.

12. 7

$40 + 3n \leq 63$; $n \leq 23/3$. The greatest whole number is 7. 8 participants would cost \$64, over the limit.

Practice worked solutions

13. 11

$25 + 6n \leq 96$; $n \leq 71/6$. The greatest whole number is 11. 12 participants would cost \$97, over the limit.

14. 11

$40 + 3n \leq 75$; $n \leq 35/3$. The greatest whole number is 11. 12 participants would cost \$76, over the limit.

15. 11

$45 + 9n \leq 152$; $n \leq 107/9$. The greatest whole number is 11. 12 participants would cost \$153, over the limit.

16. 10

$30 + 7n \leq 106$; $n \leq 76/7$. The greatest whole number is 10. 11 participants would cost \$107, over the limit.

Transfer worked solutions

1. 10

$15 + 2n \leq 36$; $n \leq 21/2$. The greatest whole number is 10. 11 participants would cost \$37, over the limit.

2. 13

$20 + 7n \leq 117$; $n \leq 97/7$. The greatest whole number is 13. 14 participants would cost \$118, over the limit.

3. 13

$40 + 7n \leq 137$; $n \leq 97/7$. The greatest whole number is 13. 14 participants would cost \$138, over the limit.

4. 11

$15 + 2n \leq 38$; $n \leq 23/2$. The greatest whole number is 11. 12 participants would cost \$39, over the limit.

Transfer worked solutions

5. 9

$40 + 3n \leq 69$; $n \leq 29/3$. The greatest whole number is 9. 10 participants would cost \$70, over the limit.

6. 11

$40 + 5n \leq 99$; $n \leq 59/5$. The greatest whole number is 11. 12 participants would cost \$100, over the limit.

Exit-ticket worked solutions

1. 12

$40 + 7n \leq 130$; $n \leq 90/7$. The greatest whole number is 12. 13 participants would cost \$131, over the limit.

2. 9

$45 + 9n \leq 134$; $n \leq 89/9$. The greatest whole number is 9. 10 participants would cost \$135, over the limit.

3. 10

$30 + 7n \leq 106$; $n \leq 76/7$. The greatest whole number is 10. 11 participants would cost \$107, over the limit.

4. 12

$30 + 4n \leq 81$; $n \leq 51/4$. The greatest whole number is 12. 13 participants would cost \$82, over the limit.

Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Represent a constraint and identify feasible values. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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